

Three Roads

Choosing a Technology Stack for My Developer Portfolio

Name	Leon Infant B
Track	General AI Fluency
Week	Week 4
Assignment	Three Roads: Choose Your Stack with AI

1. My Four Constraints

1. Free Only

I want to use free tools and free hosting options for my portfolio. I do not want to pay for hosting, deployment, or other services while building and maintaining my portfolio.

2. My Honest Skill Level

I am a BE Computer Science student with experience in HTML, CSS, JavaScript, React, Next.js, Tailwind CSS, GitHub, and basic full-stack development. I am also learning Python backend development and have worked with FastAPI. I am more comfortable with frontend development than backend development.

3. My Portfolio Sitemap and Content Map

My portfolio will have the following main sections:

- **Home** — A short introduction, my name, role, and a brief overview of my skills.
- **About Me** — My education, background, interests, and career goals.
- **Skills** — Programming languages, frontend technologies, backend technologies, databases, tools, and other technical skills.
- **Projects** — A collection of my academic, personal, and full-stack projects, including project overview, screenshots, technologies used, key features, my contribution, GitHub repository, and live demo when available.
- **Experience** — Internship experience and practical development experience.
- **Education** — My BE Computer Science education and academic background.
- **Certifications and Achievements** — Certifications, hackathons, awards, and other achievements.
- **Contact** — Contact information and links to my professional profiles.

Content I need to display: project screenshots and image galleries, project descriptions, technical skills, technologies used per project, GitHub repository links, live project demos, detailed project case studies, internship and experience, and education and achievements.

The main goal is to help recruiters and visitors quickly understand who I am, what I have built, what technologies I know, and how I approach software development.

4. How My Work Needs to Be Displayed

My portfolio needs to display my work visually and clearly. I want to use project screenshots and image galleries to show the interfaces I have designed and developed. Each project should have its own detailed section with a description, technologies used, features, GitHub repository, and live demo where available.

For larger projects, I want to include long-form project case studies explaining the problem, solution, technology stack, key features, and my contribution. The portfolio should also provide easy access to my GitHub repositories and live demos so that recruiters can explore my work directly.

2. Dynamic and Backend Requirements

At this stage, my portfolio does not require a backend or dynamic functionality. The main purpose is to showcase my projects, skills, screenshots, GitHub repositories, live demos, and project case studies.

I may add dynamic features in the future, such as a contact form, blog, CMS, database-driven project management, or other interactive features. However, I do not need these features for the first version of my portfolio.

3. My Goal

My goal is to choose a technology stack that is free, realistic for my current skill level, easy for me to maintain, and capable of presenting my projects professionally. I want to compare different approaches before making my final decision instead of choosing a stack only because it is popular or powerful.

Option 1 — React + Vite

SIMPLEST

Stack	React, Vite, Tailwind CSS, GitHub Pages
Build	Static portfolio with React components and local project data.
Hosting	GitHub Pages (Free)
Backend	Not required
Features	Responsive design, project pages, image galleries, GitHub & live demo links.
Advantages	Easy to build, fast, free, low maintenance.
Disadvantages	Limited SEO and no built-in image optimization.
Maintenance	Update project files and push to GitHub.
Display	Excellent for screenshots, galleries, case studies, and project details.

Option 2 — Next.js + Tailwind CSS

BALANCED · CHOSEN

Stack	Next.js, Tailwind CSS, Vercel
Build	Static multi-page portfolio with dedicated project pages.
Hosting	Vercel (Free)
Backend	Not required
Features	SEO, optimized images, project case studies, blog support later.
Advantages	Better performance, SEO, and scalability.
Disadvantages	More complex than React.
Maintenance	Add new pages/content and redeploy automatically.
Display	Excellent for detailed projects, screenshots, GitHub links, and live demos.

Option 3 — Next.js + Headless CMS

MOST POWERFUL

Stack	Next.js, Tailwind CSS, Headless CMS (Sanity/Contentful), Vercel
Build	Portfolio with content managed through a CMS.
Hosting	Vercel + Free CMS
Backend	No custom backend required.
Features	Editable projects, blogs, rich case studies, search, image galleries.
Advantages	Highly scalable and easy to update content.
Disadvantages	Higher learning curve and more setup.
Maintenance	Manage content in the CMS and update dependencies.
Display	Best for large portfolios with detailed case studies and rich media.

4. Pressure-Test My Front-Runner

Front-Runner: Option 2 — Next.js + Tailwind CSS

After comparing the three options, I am considering Next.js with Tailwind CSS, GitHub, and Vercel as my front-runner stack.

Q. What breaks if I pick the simplest option?

If I choose React + Vite, nothing major will break. I can still showcase my projects, screenshots, image galleries, GitHub repositories, live demos, and case studies. However, I may have fewer built-in advantages for SEO and image optimization. If my portfolio grows, I may need additional tools or more manual work to improve these areas.

Q. What do I maintain if I pick the most powerful option?

If I choose Next.js with a Headless CMS, I would need to maintain both the Next.js website and the CMS. I would also need to manage CMS content, images, dependencies, and free-tier limits. The CMS would make content updates easier, but the overall setup would be more complex than I need for my first portfolio.

Q. Can I finish in two weeks?

Yes. I believe I can finish the Next.js portfolio within two weeks because I already have experience with React, Next.js, Tailwind CSS, and GitHub. I can first build the essential pages and project case studies and add advanced features later.

Q. Does it show my work the way it needs to be shown?

Yes. Next.js can effectively display my project screenshots, image galleries, GitHub repositories, live demos, technical skills, internship experience, education, certifications, and long-form project case studies. The portfolio can have dedicated project pages, which will make my work easy for recruiters and visitors to explore.

Q. Do I need a backend right now?

No. I do not need a backend for the first version of my portfolio. My main goal is to showcase my work. A static Next.js portfolio is enough for this purpose. If I later need features such as a database, authentication, CMS, or other dynamic functionality, I can add them later.

Q. Can I maintain this?

Yes, I believe I can maintain this stack. I already have experience with React, Next.js, Tailwind CSS, and GitHub. I will need to update dependencies and add new projects as I complete them, but the overall maintenance should be manageable.

PRESSURE-TEST RESULT

After pressure-testing the three options, I believe the Next.js + Tailwind CSS option is the best balance for my current needs. React + Vite is simpler and easier to maintain, but Next.js provides better SEO, image optimization, and room for future expansion. Next.js + Headless CMS is more powerful, but it introduces additional complexity and maintenance that I do not need at this stage. The Next.js static approach allows me to finish within two weeks, maintain the portfolio myself, and display my work professionally without needing a backend.

5. Final Decision and Rationale

FINAL STACK

Next.js + Tailwind CSS, hosted on Vercel

I chose this stack because it matches my current skills and the way I want to present my work. I already have experience with React, Next.js, Tailwind CSS, and GitHub, so I believe I can build and maintain the portfolio myself. It also gives me a professional way to showcase my project screenshots, image galleries, GitHub repositories, live demos, and long-form project case studies.

Why not React + Vite: I considered it because it is the simplest option and would be easy to build and maintain. However, I decided not to choose it because I want better SEO, image optimization, and more flexibility for future improvements.

Why not Next.js + Headless CMS: I also considered it because it is the most powerful and scalable option. However, I decided not to choose it for my first version because it adds more complexity and maintenance than I currently need. I do not need a CMS or backend yet because my portfolio is mainly for showcasing my work.

Free and achievable: The chosen stack is free to use with the available free hosting plan and does not require a backend for the first

version. I believe I can finish the portfolio within two weeks because I already know the main technologies involved.

Two guiding questions: Most importantly, I asked myself two questions — *Can I maintain this?* Yes, because I already have experience with the technologies and the project structure is manageable. *Does it show my work well?* Yes, because Next.js allows me to create professional project pages with screenshots, galleries, GitHub links, live demos, and detailed case studies.

Overall, I chose Next.js with Tailwind CSS because it gives me the best balance between simplicity, professional presentation, maintainability, and future growth without adding unnecessary complexity.